



SWEESH

Every line of a resume, **checkable.**

Sweesh is a chain of custody for facts about a person — portable as one badge, verifiable end to end, revealing nothing beyond the answer.

THE PROBLEM — WHAT WE HAVE TODAY

Today, a career is an unverified document.

An employer takes it on faith — or pays a screening firm to phone it in by hand.

Nothing here is signed.

Every line is self-asserted. No registrar, no HR system, no manager has vouched for any of it.

The proof already exists.

It sits in a registrar's transcript, an HRIS record, an old manager's memory — disconnected, never asked.

It all forgets you.

Each system loses interest in you the day you leave — which is exactly the day you need it to answer.

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EDUCATION

M.S., Computer Science, New York University, Courant Institute of Mathematical Sciences

Aug. 2026 – May 2028 (Expected)

Focus: Machine Learning, Large Language Models, AI Systems.

B.E., Computer Science, BITS-Pilani, Pilani Campus

2020 – 2024

Coursework: Data Structures & Algorithms, Operating Systems, Compilers, Computer Networks, Databases, Computer Architecture.

GPA: 8.11/10

EXPERIENCE

Software Engineer, GenAI / LLM Systems

Qualcomm

July 2024 – Aug 2026

Hyderabad, India

- Architected **Code Buddy**, a multi-agent GenAI code-review platform deployed across **4,000+** repos for **12,000+** engineers, running detect-then-fix agents under a tool-calling orchestrator that generate self-validating fixes and clear a static-analysis guardrail before posting.
- Scaled the platform to **2M+** automated code reviews (**180,000+** monthly), saving an estimated **114,000+** engineers-hours per month.
- Cut review-retrieval latency via a Tree-sitter symbolic-retrieval layer over a ccls/ctags dependency graph with TTL symbol caching.
- Benchmarked agent precision/recall on golden datasets with prompt-evaluation harnesses, enabling safe iteration across model upgrades.
- Automated interactive 4G/5G NAS protocol testing with an agentic tool that reasons over C/C++ / modems source through a **14-tool** calling loop.
- Built an autonomous LLM agent that triages NAS modem hotline escalations end-to-end via a **13-tool** calling loop: it parses issue emails, searches Perfcore source and CR/JIRA databases, and shelves a 3GPP-cited candidate fix after a **4-model** consensus review.

Software Development Intern, Performance / Systems

Advanced Micro Devices (AMD)

Jan. 2024 – Jun. 2024

Bangalore, India

- Cut bus-trace analysis runtime from a **week to 45 minutes** by building a multi-threaded LSF pipeline that streamed and sharded **250 GB+** datasets.
- Automated triage of **1,000+** JIRA issues via the REST API with hierarchical linkages and auto-summaries.
- Drove hardware-selection decisions by benchmarking CPUs and servers with Sysbench and SPEC CPU 2017 on throughput and tail latency.
- Authored reusable benchmark suites and automation scripts that standardized the team's performance-analysis workflow across platforms.

Software Development Intern

Standard Chartered CBIS Pvt Ltd

May 2023 – Jul. 2023

Chennai, India

- Achieved 98.61% line coverage on the SC Pay backend by writing 100+ JUnit and Mockito tests.
- Earned the **CCIB SCPay 2023 Award** for migrating 10+ repos from Jenkins to Azure DevOps with CI/CD.

ACHIEVEMENTS

- Won SparQ Qualcomm TechFest 2026**: presented the CabinAI architecture to the VP of AI and VP of Automotive at Qualcomm.
- SparQ Qualcomm TechFest 2025 Finalist**: presented the Code Buddy multi-agent architecture to the Qualcomm CTO.
- DAAD Scholarship**, German Academic Exchange Service: selected for Mixed Reality research at TU Braunschweig, Germany.
- CCIB SCPay 2023 Award**, Standard Chartered Bank, for the Jenkins to Azure DevOps migration.
- JumpStart Hackathon**, Publicis Sapient: top performance, earned exclusive program access.
- JEE Advanced**: All-India Rank **2358** / **1.2M+** (top 0.2%), Indian Institutes of Technology.

SELECTED PROJECTS

CabinAI, Hybrid Multi-Agent VLA for Automotive In-Cabin Intelligence

2026

- Architected a **T-agent hybrid edge-cloud vision-language-action system** for real-time driver monitoring: 30 FPS MediaPipe face-mesh DMS and Qwen-7B INT4 (**40 tok/s**) handle safety and voice on-device, escalating complex reasoning to Qwen3-VL-32B / gpt-oss-20b on Qualcomm AIC100.
- Routed **80%+** of queries to edge via a sub-1 ms rule-based router, with multi-tier fallback (AIC100, QGenie Claude, mock) keeping every agent online.
- Built a **15-minute-ahead** proactive fatigue forecast at **500 ms** by sending a **13.5 KB** INT5 buffer plus the latest cabin frame to a Qwen3-VL-32B VLM.
- Powered local RAG over a 31-document corpus in **<50 ms** via BGE-smal ONNX (no PyTorch); preempts non-safety bus events when drowsiness > 0.7.

MyAI, Privacy-First Personal On-Device AI Assistant

2026

- Designed a **4-tier on-device inference router** (trie, cache, tools, INT4 LLM) that resolves most queries without loading the full LLM.
- Ran Qwen2.5-1.5B / Llama-3.2-3B INT4 at **~30 tok/s** (w/lat6) on Hexagon NPU v79 via the Qualcomm Genie SDK and QNN runtime.
- Enforced **kernel-level privacy** by omitting the INTERNET permission, gated by a Gradle CI task (assertNoInternet) proving zero network egress.
- Exposed an on-device agent / tool layer (calculator, datetime, calendar, contacts, alarm) that the LLM invokes via structured JSON tool-calls.
- Added a feedback-driven learning cache that, after one thumbs-up, serves a repeat query in **<10 ms** instead of **~600 ms** of model inference.

HealthSense, On-Wearable Multimodal Biosignal Foundation Model

2026

- Trained **PanLUNA**, a **5.4M-param** multimodal Transformer foundation model that fuses five wearable biosignals (EEG, ECG, PPG, EMG, IMU) into a shared 1024-l latent, then routes it to 4 task heads detecting cardiac arrhythmia, sleep stage, stress, and EEG abnormality from one forward pass.
- Ran INT5 inference on Hexagon NPU at **~95 tok/s** / **1B pd** per pass with a one-line ONNX hot-swap between real and synthetic cardiac models.
- Validated the cardiac head on **PhysioNet MIT-BIH AFB** (530 windows, 4 records): **96.8%** AFB sensitivity, **91.7%** sinus specificity.
- Built **BioTrain**, an on-device fine-tuning tool that personalises the model to each wearer (last-2-block backprop + GroupNorm in 0.67 MB RAM), lifting stress-detection F1 from **0.78 to 0.91** in **~3.6 s** under **50 mW** while keeping all raw biosignals on the device.

Compiler Construction (C) C

Feb. 2023 – Apr. 2023

- Academic Project
- Built an end-to-end compiler (DFA twin-buffer lexer, LL(1) parser with FIRST/FOLLOW gen, AST, codegen) cutting parse-tree memory by **70%**.
- Added semantic analysis with type checking and scope resolution, plus panic-mode error recovery for graceful handling of malformed input.

LEADERSHIP

- Founding President**, Kallipatnapu AB/VR Lab, BITS Pilani (2023 – 2024): led a 40-student team and ran a VR experience zone earning **40k**.
- Blockchain Dept Coordinator**, BITS Coding Club (2022 – 2023): chartered the new department, ran recruitments, organized an **ICP Hackathon**.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Kotlin, JavaScript, SQL, Bash, C#.

ML / LLM Stack: PyTorch, Hugging Face Transformers, LangChain, vLLM, Ollama, ONNX Runtime, RAG, LoRA / QLoRA fine-tuning, multi-agent orchestration, prompt evaluation, structured-output / tool use, embeddings (Tree-sitter, BGE, ChromaDB).

Edge / On-Device AI & Systems: Snapdragon AIC100, Hexagon NPU, Qualcomm Genie SDK + QNN, AI Hub, INT4/INT5 quantization, Llama.cpp; Linux, Docker, Kubernetes, Redis, Flask, Azure DevOps, Git, CI/CD.

Verification is slow, manual, and expensive.

And the cost sits on the party who captures none of the value.

\$30–100

per candidate

background screening

~3 days

of phone calls

registrar · HR · old manager

\$20–65

per check

employment / income verification

The employer wants the answer. The registrar or HR team produces it, gets paid nothing, and carries the liability. That asymmetry — not a missing technology — is why this has stayed broken.

Four gaps define our opening.

Payroll knows dates, not contribution

Payroll systems record titles and dates, nothing about what you did.

Manager attestations under delegated authority

Credentials verify, issuers do not

Blockcerts checks the document — not that the issuer is a real, accredited institution.

Recursive authority chains to a verifier-chosen root

Verification costs privacy

Answering a query means revealing the record — and callbacks tell the issuer where you applied.

Selective disclosure + zero-knowledge path proofs

Records die with the relationship

Your work email stops working the day you leave — the day you need it.

Issue once at the moment of truth, prove for life

THE ANSWER

No resume. One badge.

Every fact is signed once, **at the moment it becomes true** — by the only party who could have known it.

The person collects those statements and sends one link carrying only what they choose to reveal. No callback. No phone call. No resume.

1

Issue at the moment of truth

The university signs the degree when it confers it.
HR signs the dates at hire and at departure.

2

Verify the signers, not just the signature

A signature means nothing unless you know who signed. Sweesh records authority the same way it records facts.

3

Reveal nothing beyond the answer

The employer learns the claim is backed by a root they trust — not who signed it, or anything not deliberately sent.

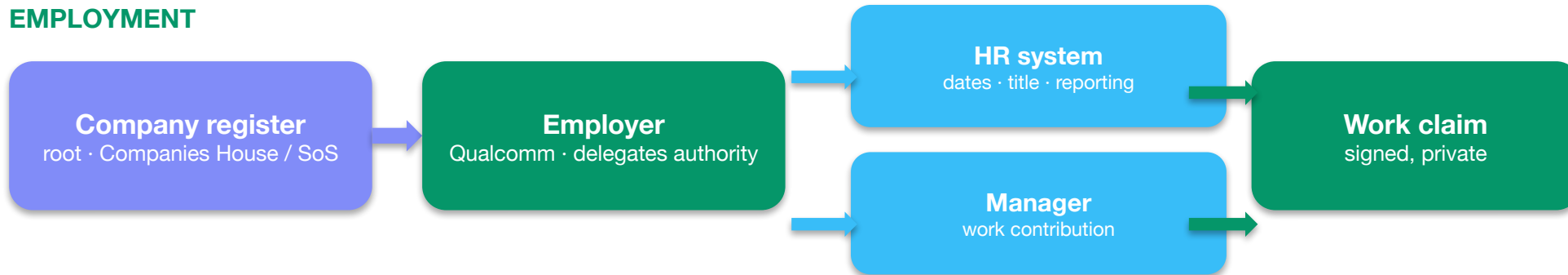
Every claim walks to a root the verifier chose.

Each arrow is an accreditation. If any link is missing, unverifiable, expired, or revoked — the claim does not pass.

EDUCATION



EMPLOYMENT



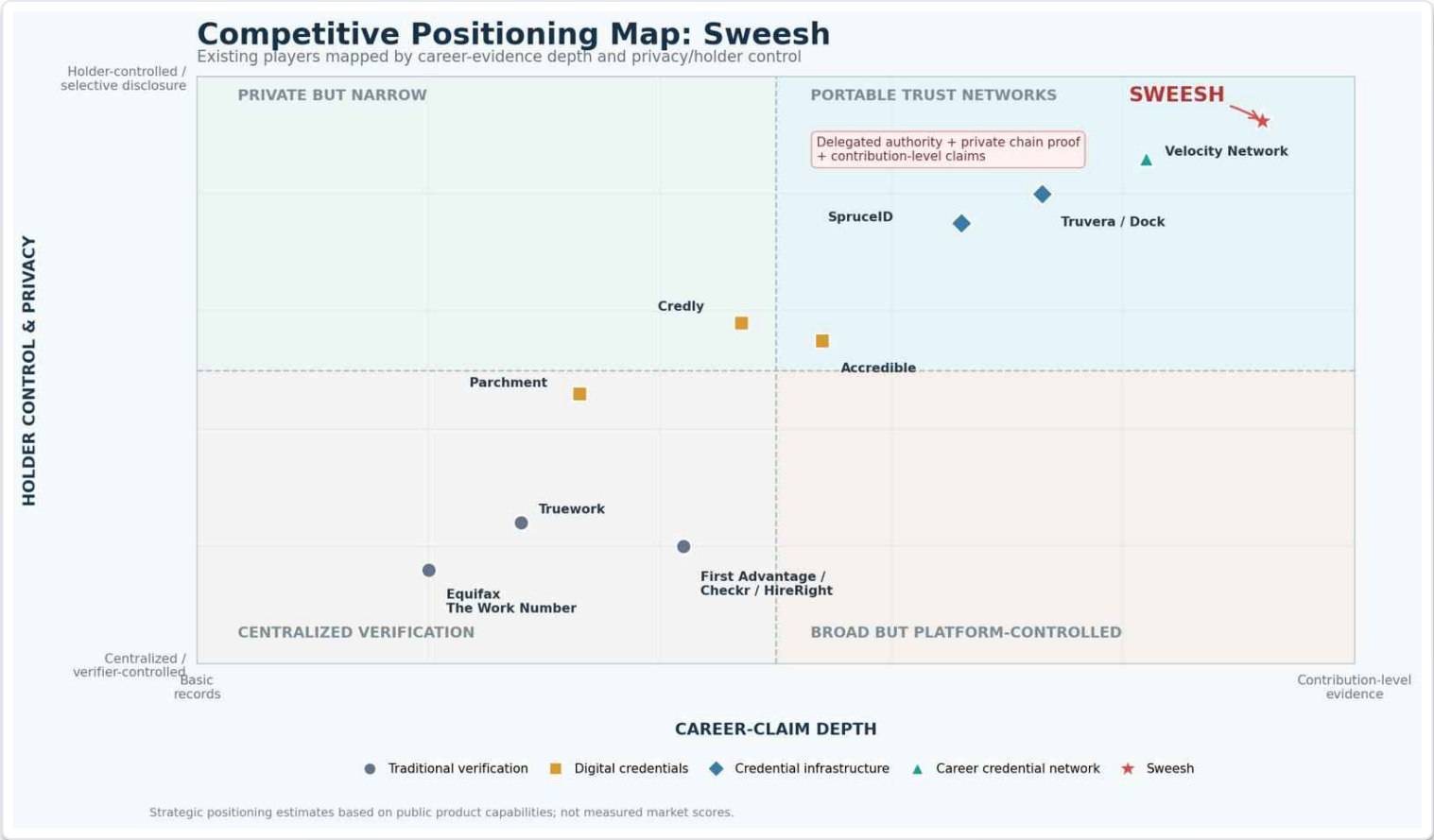
Scope only ever narrows. HR grants the manager authority but never sees what the manager writes — so your old employer cannot read what your manager said about you.

Who verified what.



We sit where nobody else is.

Holder control & privacy on one axis, depth of career evidence on the other.



The portable-trust corner. Delegated authority + private chain proof + contribution-level claims — the only player with all three.

A large, growing market.

\$24B+

adjacent market today

Background screening \$14B + digital identity \$10B

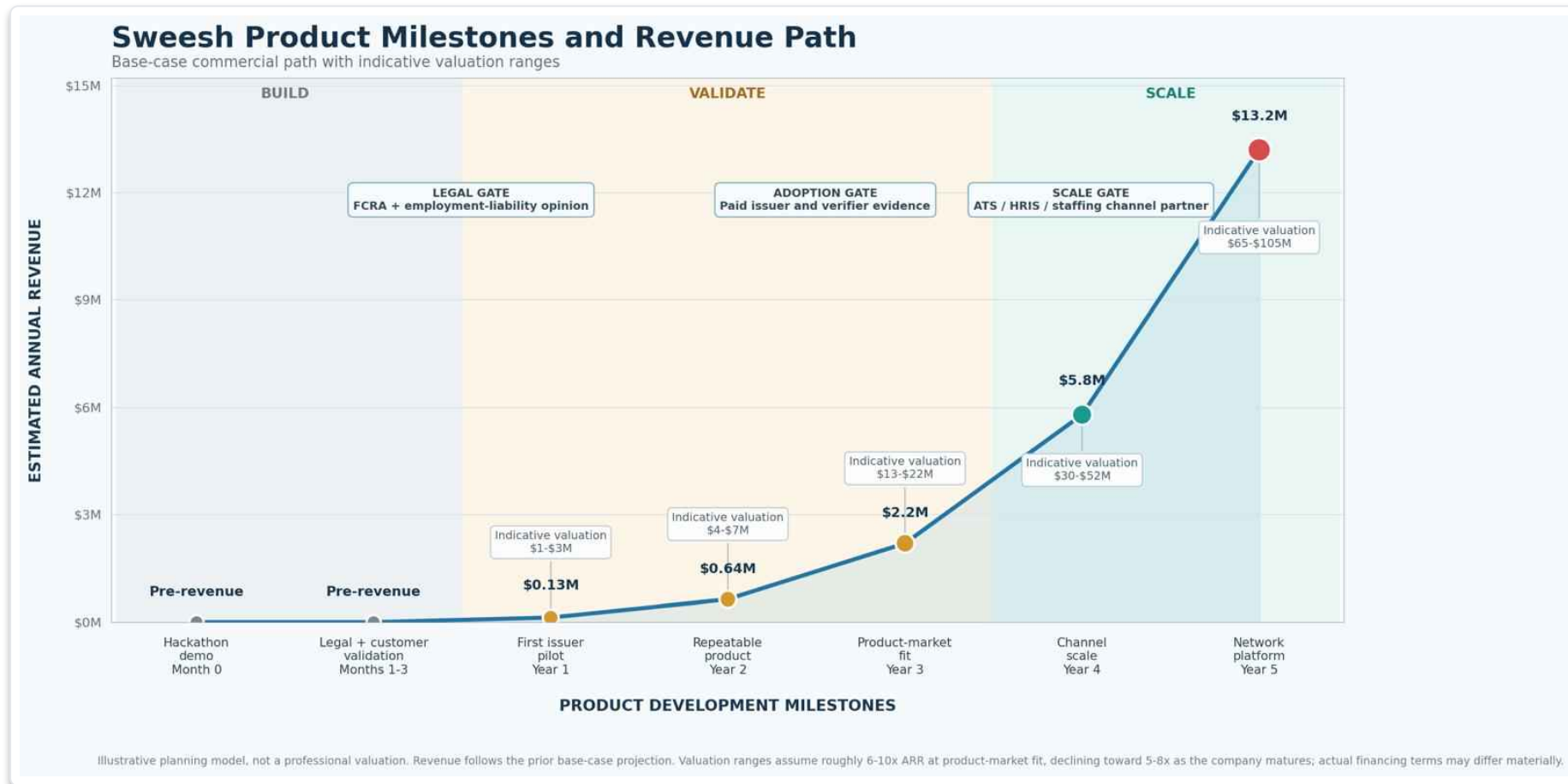
\$43–48B

by 2031

screening +6–7%/yr · digital identity +15–18%/yr

Group	Players	Provide	Pay
Employment verification	Equifax, Truework	Dates, title, income	\$20–65 / check
Background screening	First Advantage, Checkr	Criminal, employment, edu	\$30–100 / candidate
Digital credentials	Credly, Accredible	Diplomas, badges	Issuer subscription
Career networks	Velocity Network	Career wallet	Network / tx fees
Credential infra	SpruceID, Truvera	Issuance, wallets, APIs	Licence + API

From hackathon demo to \$13.2M.



Base case: \$13M in year five (450 issuers, 200 employer/API customers) · indicative valuation \$65–105M. Achievable only with a distribution partner — ATS, HRIS, or staffing platform.

Free for candidates. Issuers and verifiers pay.

Basic credential issuer	Free – \$1.5k / yr
HR / university issuer platform	\$6k – \$15k / yr
Enterprise issuer · HRIS / SIS integration	\$25k – \$75k / yr
Employer verification platform	\$12k – \$60k / yr
Individual badge verification	\$2 – \$8 / badge
ATS / background-screening API partner	\$50k – \$250k / yr + usage

Recurring subscriptions + verification fees. Not a pure per-check model — reusable credentials naturally reduce repeated checks.

We don't out-cover the incumbent. We out-scope it.

The incumbent: Equifax Work Number

839M+ employee records, fed automatically through payroll, access sold to verifiers. A copy of most of the American workforce's history.

What payroll structurally cannot do

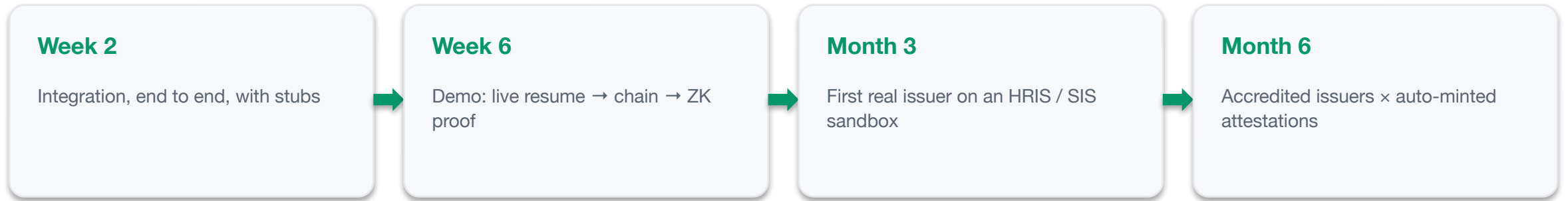
Granular work claims · candidate-controlled disclosure · cross-border · verification without paying per check.

Cold start, answered.

Sell verification-request deflection to registrars and HR teams who field these requests for free today. Credentials enter the network as exhaust — the party who acts first finally captures value.

The honest risk: most employers restrict manager references to dates and title. Our design makes evaluation structurally unrepresentable — factual, scoped attestations with the company as legal attester.

Six weeks to demo. Then the metric that matters.



The metric that matters: accredited issuers × attestations auto-minted per month with zero human touch — and whether that number grows without sales touching it.

Team A — Protocol & core

3–4 engineers · cryptography

Attestations, delegated authority, ZK chain proofs.

Team B — Data & graph

2–3 · data science

Real institution graph from ORCID, ROR, DAPIP; edge weighting.

Team C — Atlas & demo

2–3 · frontend / WebGL

The living map + badge inspector that carry the pitch.



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